

Create PCBs up to 16 Layers with KiCad

Operational on Apple OS X, Linux and Windows, KiCad is an electronic design automation software tool to design and generate professional schematics and printed circuit boards. A well-known software tool, KiCad is released under the open source General Public Licence v2 (GNU GPL). Try out KiCad included in this month's DVD exclusively available with EFY Plus

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Designed and written by Jean-Pierre Charras, and under active development by the KiCad Developers Team, KiCad features an integrated environment that allows you to create schematic diagrams and PCBs up to 16 layers.

KiCad can be implemented and interoperated on multiple computer platforms. It is an open source software suite for EDA and it facilitates the design of schematics for electronic circuits and their conversion to PCB designs as well. Basically, it provides the user with an integrated environment for PCB layout design and schematic capture.

Tools exist within the package to create a bill of materials (BOM), artwork and Gerber files and 3D views of the PCB and its components too. A number of component libraries are available with the ability to add custom components by the users.

Interestingly, there are tools avail-

KiCad is considered mature and can be used for the successful development and maintenance of complex electronic boards. KiCad does not present any board-size limitation, and it can easily handle up to 16 copper layers and 12 technical layers. KiCad can also create all the files necessary for building printed boards like Gerber files for photo-plotters, drilling files, component location files and a lot more.

able to assist with importing components from other EDA tools like Eagle, for example. With multiple language support and 3D PCB viewing, there is a built-in basic autorouter as well.

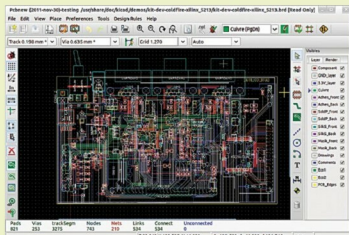
Features and functionality in three steps

KiCad allows you to develop your printed circuit board (PCB) via three

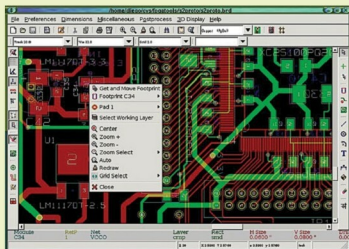
interconnected and independent main applications: Eeschema, Cvp pcb, Pcbnew. This is simply done in three steps.

Step 1: Schematic capture. With the KiCad schematic editor Eeschema, you can create a sophisticated electronic sheet or a group of hierarchical sheets. Several schematic components come with the default KiCad library. An Electrical Rules Check (ERC) tool is available too.

Step 2: Component association. Cvp pcb allows you to associate



KiCad main window



KiCad Pcbnew